

Dear Editor,

We submit our manuscript, “Adversarial Vulnerability of Machine Learning GNSS Spoofing Detectors to Transmitted Spoofing Signals,” for consideration as an Original Article in *Satellite Navigation*.

The problem the manuscript addresses

Machine learning spoofing detectors are moving from research prototypes towards integration in receivers, and the accuracies reported for them are high. Those accuracies are obtained against fixed corpora of recorded attacks. A recorded attack does not respond to the detector being tested, so what such an evaluation establishes is performance against the attacks someone else already chose to record. A deployed receiver faces an adversary who selects what to transmit.

Closing that gap requires the attack to be a signal. Work that perturbs receiver observables directly must argue separately that the perturbation corresponds to something a transmitter could produce, and the argument is difficult to make because several observables are computed from the same correlator accumulations within one integration period: displacing one while holding the others fixed specifies a receiver state no incident waveform produces.

What we did

Every attack in this study is constructed as a radio waveform. A counterfeit GPS L1 C/A signal is synthesised at sample level, summed with authentic recordings, passed through a model of the receiver front end and tracked by an open software receiver, so the observables each detector reads are produced by receiver processing of a transmittable signal. Realizability is settled by the construction of the experiment, not by an argument attached to it. The generator is validated by reproducing a recorded attack whose construction is published.

We sweep 209 transmitter settings spanning counterfeit power, code pull-off rate and carrier alignment, of which 126 satisfy the published conditions for receiver takeover, and measure fourteen detectors, eight classical and six deep, against that subset. Every detector is held at one common operating point, and the false alarm rate each pays is reported beside every result, so no apparent robustness is credited to a displaced threshold.

What we found

Clean detection performance is a poor predictor of resistance to transmitted attacks, and near the top of the ranking the two orderings run close to inverted. The detector with the highest clean F_1 and the lowest false alarm rate fails to flag 55 of the 126 attacks. Where resistance does appear it is purchased with false alarm rates of 46 to 74 percent on authentic data, which is not operationally available to a receiver. And detection failure is not confined to the classifier: following one missed attack through the navigation stage, the receiver clock is driven at 48.89 nanoseconds per second against a commanded 48.87.

Why *Satellite Navigation*

The manuscript addresses resilient positioning, navigation and timing, a core topic of the journal, and connects directly to the literature on learning based GNSS spoofing detection that the journal has published. Its findings are stated in the terms a receiver designer must weigh, namely detection against false alarms and the consequence of a miss for the position and timing solution. The library of transmitted attacks is a reusable artefact for evaluating physics constrained and certified detectors, which we expect to be of direct use to the journal's readership.

Declarations

This manuscript has not been published and is not under consideration for publication elsewhere, in whole or in part. All authors have seen and approved the manuscript in its current form and agree to its submission to *Satellite Navigation*. The authors declare that they have no competing interests, and we are not aware of any policy issues relevant to this submission. This submission is not intended for a special issue.

Thank you for considering our manuscript.

Sincerely,

Wasiu Akande Ahmed and Joel Segun Ojerinde (corresponding authors)

On behalf of all authors

Regional Centre for Space Science and Technology Education in Asia and the Pacific (RCSSTEAP),
Hangzhou International Innovation Institute, Beihang University, Hangzhou 311115, China
ahmedwasiu24@buaa.edu.cn; joelojerinde@buaa.edu.cn